



MIST- MEDICAL INSTITUTE FOR SCREENING TEST

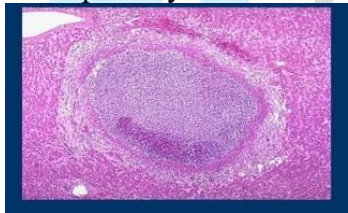
PATHOLOGY SUBJECT WISE TEST DATED ON 24TH APRIL 2026

QUESTIONS

Q1. A 58-year-old man with long-standing hypertension develops sudden chest pain and collapses. ECG shows ST elevation in anterior leads. Coronary angiography confirms complete occlusion of the left anterior descending artery. After 30 minutes of ischemia, myocardial cells show cellular swelling and reversible mitochondrial dysfunction. Which of the following mechanisms is the earliest biochemical event responsible for reversible cell injury in this patient?

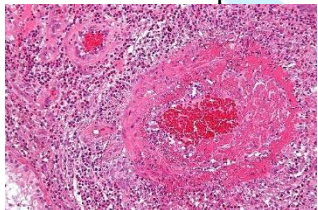
- a) Increased lysosomal membrane permeability
- b) ATP depletion due to oxidative phosphorylation failure
- c) Activation of caspases
- d) Nuclear chromatin fragmentation

Q2. A 45-year-old male with a history of chronic alcohol use disorder is brought to the clinic by his brother. The patient has been experiencing a high-grade fever, night sweats, and a persistent cough for the last ten days. His brother notes that the patient "coughed up something that smelled like rotting garbage" earlier that morning. Physical examination reveals decreased breath sounds and dullness to percussion over the right lower lung field. A CT scan of the chest identifies a 4 cm thick-walled cavity containing an air-fluid level in the superior segment of the right lower lobe. A biopsy of the cavity wall is performed, revealing a localized collection of neutrophils, cellular debris, and complete enzymatic transformation of the pulmonary parenchyma into a viscous, liquid mass. Which type of necrosis is the primary mechanism behind the formation of this lung abscess?



- a) Coagulative necrosis
- b) Liquefactive necrosis
- c) Fat necrosis
- d) Fibrinoid necrosis

Q3. A 28-year-old woman with a five-year history of Systemic Lupus Erythematosus (SLE) presents to the emergency department with a new-onset, non-blanching purpuric rash on her lower extremities and worsening proteinuria. Her current flare has been characterized by intense joint pain and malar rash. A skin biopsy of a representative lesion is performed. Microscopic examination of the small dermal arteries reveals heavy inflammation and circumferential vessel wall destruction. The walls appear thickened by an accumulation of smudgy, eosinophilic (bright pink), proteinaceous material that resembles clotted blood. Which of the following types of necrosis is the hallmark of this patient's vascular pathology?



- a) Fat necrosis
- b) Fibrinoid necrosis
- c) Liquefactive necrosis
- d) Caseous necrosis



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Q4. A 54-year-old male with a long-standing history of chronic gastroesophageal reflux disease undergoes an endoscopy, which reveals salmon-pink, velvety patches in the lower esophagus. A biopsy of this region is performed to confirm the diagnosis. What is the characteristic microscopic finding required to confirm Barrett's esophagus?

- a) Squamous metaplasia of the columnar epithelium
- b) Presence of goblet cells within the columnar epithelium
- c) Hyperplasia of the basal layer of squamous epithelium
- d) Dysplastic changes in the superficial squamous layer

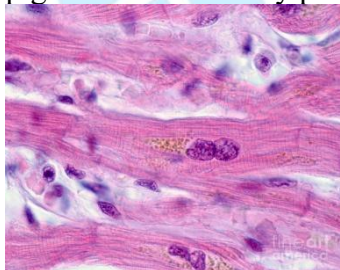
Q5. A 62-year-old male is brought to the emergency department with severe, crushing chest pain radiating to his left jaw. Despite resuscitation efforts, the patient passes away. Autopsy and subsequent electron microscopy of the myocardium reveal profound swelling of the organelles, rupture of the sarcolemma, and the presence of large, opaque, proteinaceous aggregates within the mitochondrial matrix. Which of the following ultrastructural findings, indicative of irreversible cell injury, is described in this case?

- a) Mitochondrial swelling
- b) Ribosomal detachment
- c) Amorphous flocculent densities
- d) Nuclear chromatin clumping

Q6. A 52-year-old woman presents with new-onset seizures and a persistent headache. A brain MRI reveals a well-circumscribed, slow-growing mass attached to the dura mater. Histological evaluation shows whorled patterns of cells and numerous laminated, concentric calcifications. Given the patient's normal serum calcium levels, which of the following is the most likely diagnosis?

- a) Medullary carcinoma of the thyroid
- b) Papillary serous cystadenocarcinoma
- c) Papillary thyroid carcinoma
- d) Meningioma

Q7. A 75-year-old elderly patient undergoes myocardial biopsy during evaluation of heart failure. Histology reveals yellow-brown perinuclear granules within cardiomyocytes, representing wear-and-tear pigment. Which pigment is most likely present?



- a) Melanin
- b) Lipofuscin
- c) Hemosiderin
- d) Bilirubin

Q8. A 28-year-old male presents to the ICU with high-grade fever, hypotension, and altered mental status following a deep soft-tissue infection. Blood cultures grow *Staphylococcus aureus*. Molecular analysis of his macrophages reveals the assembly of the NLRP3 inflammasome, which activates Caspase-1, leading to a form of programmed cell death characterized by cell swelling and membrane rupture. Which pair of pro-inflammatory cytokines is released into the systemic circulation as a result of this specific cell death pathway?

- a) IL-2 and IL-4



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- b) IL-10 and IL-13
- c) IL-1 β and IL-18
- d) TNF- α and IFN- γ

Q9. A 45-year-old male with poorly controlled diabetes presents with a painful, fluctuant swelling on his right forearm. Laboratory results show a significant shift to the left (increased bands). At the site of infection, chemical mediators like IL-8 and Leukotriene B₄ have created a gradient. The neutrophils begin to squeeze through the intercellular junctions of the vascular endothelium to reach the interstitial space. Which specific step of the inflammatory cascade and associated molecule is primarily responsible for this "squeezing" motion?

- a) Margination mediated by Sialyl-Lewis X
- b) Diapedesis mediated by PECAM-1
- c) Rolling mediated by P-selectin
- d) Opsonization mediated by C3b

Q10. A 28-year-old woman is diagnosed with lobar pneumonia. As the inflammatory response intensifies, cytokines like TNF and IL-1 act on the local pulmonary capillary endothelium. This causes the transient and weak binding of leukocytes to the vessel wall, resulting in a "tumbling" or "rolling" motion along the surface. Deficiency in which of the following molecules would most likely impair this specific initial step of leukocyte recruitment?

- a) Selectins
- b) Integrins
- c) C5a and C3a
- d) NADPH Oxidase

Q11. A 32-year-old woman with pulmonary tuberculosis undergoes lung biopsy showing granuloma formation with epithelioid cells and Langhans giant cells. Which cytokine is primarily responsible for macrophage activation in granulomatous inflammation?

- a) Interleukin-4
- b) Interferon- γ
- c) Interleukin-10
- d) Interleukin-13

Q12. A 4-year-old boy is brought to the pediatrician due to recurrent, deep-seated skin abscesses and several episodes of cervical lymphadenitis caused by *Staphylococcus aureus*. His medical history is significant for a previous bout of liver abscesses and a fungal pneumonia caused by *Aspergillus*. On examination, he has multiple scars from previous surgical drainages and significant hepatosplenomegaly. Laboratory findings reveal a persistent neutrophilia. A Dihydrorhodamine (DHR) flow cytometry test is performed, showing a lack of green fluorescence in stimulated neutrophils, confirming a failure to generate reactive oxygen species (ROS). Which of the following is the most likely underlying molecular defect?

- a) Deficiency of Myeloperoxidase (MPO)
- b) Deficiency of NADPH Oxidase
- c) Mutation in the LYST gene (Chédiak-Higashi)
- d) Deficiency of Adenosine Deaminase

Q13. A 5-year-old boy is brought to the clinic due to recurrent, severe bacterial skin infections and a recent onset of progressive numbness in his feet. On physical examination, the physician notes oculocutaneous albinism (very pale skin and light-colored irises) and horizontal nystagmus. A peripheral blood smear is ordered, which reveals neutrophils containing massive, peroxidase-positive giant granules. Which of the following is the primary molecular defect responsible for this patient's presentation?



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- a) Deficiency of NADPH oxidase
- b) Defect in lysosomal trafficking gene
- c) Deficiency of Integrin CD18
- d) Mutation in the Myeloperoxidase gene

Q14. A 24-year-old African American woman presents to the dermatology clinic concerned about a growth on her right earlobe. She notes that the lesion began as a small bump six months after she had her ears pierced, but it has since grown significantly, now extending well beyond the original site of the piercing. The lesion is firm, itchy, and occasionally painful. Physical examination reveals a thick, raised, rope-like mass of tissue. A biopsy is performed, revealing disorganized, thick "glassy" eosinophilic collagen bundles. Which of the following is the most likely diagnosis and the predominant collagen type found in this lesion?

- a) Hypertrophic scar; Type III Collagen
- b) Hypertrophic scar; Type I Collagen
- c) Keloid; Type III Collagen
- d) Keloid; Type I Collagen

Q15. A 32-year-old male presents to the clinic with painful, fluctuant swelling in his left inguinal region. He recalls having a small, painless ulcer on his penis about three weeks ago that healed spontaneously without treatment. On examination, there are multiple enlarged, matted lymph nodes above and below the inguinal ligament. A biopsy of the lymph node is performed, revealing a central area of necrosis surrounded by neutrophils and a rim of epithelioid histiocytes, forming a characteristic star-shaped pattern. Which of the following descriptions best characterizes the type of granuloma found in this patient?

- a) Non-caseating granuloma
- b) Caseating granuloma
- c) Stellate granuloma
- d) Foreign body granuloma

Q16. A 28-year-old man develops severe bacterial sepsis. Macrophages recognize lipopolysaccharide (LPS) on Gram-negative bacteria through a pattern recognition receptor, leading to cytokine release. Which receptor is primarily responsible for recognizing LPS?

- a) Toll-like receptor 4
- b) Toll-like receptor 2
- c) NOD-like receptor
- d) RIG-I receptor

Q17. A 40-year-old male is diagnosed with a high-grade nasopharyngeal carcinoma. Molecular analysis of the tumor cells reveals that the virus-infected cells have downregulated the expression of MHC Class I molecules to evade detection by Cytotoxic T-lymphocytes (CD8+). However, the patient's Natural Killer (NK) cells are still able to recognize and lyse these malignant cells. Which of the following best explains the activation of NK cells in this scenario?

- a) Binding of CD16 to the Fc portion of IgG
- b) Loss of inhibitory signaling due to the absence of MHC-I
- c) Activation of the Classical Complement pathway



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d) Recognition of tumor-specific antigens via MHC-II

Q18. A 2-year-old boy presents to the pediatrician with a history of recurrent *Pneumocystis jirovecii* pneumonia and severe sinopulmonary infections. Laboratory evaluation reveals markedly elevated Serum IgM levels, while IgG, IgA, and IgE are nearly undetectable. Flow cytometry reveals that his T-lymphocytes fail to express a specific surface protein required for effective interaction with B-lymphocytes during the germinal center reaction. A defect in which of the following molecules is the most likely cause of this patient's failure to undergo immunoglobulin class switching?

- a) CD40 Ligand
- b) Complement C3 receptor
- c) MHC Class I alpha chain
- d) Fc-gamma receptor (CD16)

Q19. A 12-year-old girl with a known history of atopic dermatitis and seasonal allergies is brought to the emergency department with an acute exacerbation of asthma. Sputum analysis reveals a high concentration of Curschmann spirals and Charcot-Leyden crystals. Her serum shows significantly elevated levels of IgE and a marked increase in the peripheral eosinophil count. Which T-helper cell subset and its associated cytokine profile are primarily responsible for orchestrating this specific immune response?

- a) Th1 cells producing IFN- γ and IL-2
- b) Th2 cells producing IL-4, IL-5, and IL-13
- c) Th17 cells producing IL-17 and IL-22
- d) Regulatory T-cells (Tregs) producing TGF- β and IL-10

Q20. A 45-year-old patient being treated for bacterial endocarditis with high-dose intravenous penicillin develops sudden-onset fatigue, jaundice, and dark-colored urine. Laboratory studies show a low hemoglobin level, elevated indirect bilirubin, and a positive Direct Antiglobulin (Coombs) test. The pathophysiology involves penicillin acting as a hapten, binding to the surface of Red Blood Cells (RBCs), which then triggers the production of IgG antibodies against the RBC-drug complex. Which type of hypersensitivity reaction is responsible for the destruction of red blood cells in this patient?

- a) Type I (Immediate) Hypersensitivity
- b) Type II (Cytotoxic) Hypersensitivity
- c) Type III (Immune Complex) Hypersensitivity
- d) Type IV (Delayed-type) Hypersensitivity

Q21. A 34-year-old healthcare worker undergoes a routine screening Mantoux (Tuberculin) skin test before starting a new position. After the intradermal injection of Purified Protein Derivative (PPD), the site is initially clear. However, 48 hours later, the patient returns with a firm, raised, and reddened area of induration measuring 18 mm at the injection site. Biopsy of the indurated area would reveal a dense perivascular accumulation of CD4⁺ T-lymphocytes and activated macrophages, with no evidence of antibody or complement deposition. Which type of hypersensitivity reaction is responsible for this delayed skin response, and what is the primary cellular mediator involved?

- a) Type I Hypersensitivity; Mast cells
- b) Type II Hypersensitivity; Natural Killer cells
- c) Type III Hypersensitivity; Neutrophils
- d) Type IV Hypersensitivity; T-lymphocytes

Q22. A 65-year-old male presents with generalized weakness, persistent lower back pain, and a weight loss of 8 kg over the past three months. Physical examination reveals a swollen, enlarged tongue (macroglossia) and periorbital ecchymosis ("raccoon eyes"). Laboratory investigations show a "punched-out" lytic lesion in the skull on X-ray and a



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serum creatinine of 2.4 mg/dL. A rectal fat pad biopsy is performed. Under Hematoxylin and Eosin (H&E) stain, there is an accumulation of extracellular, amorphous, eosinophilic material. When stained with Congo Red and viewed under polarized light, the tissue exhibits a characteristic apple-green birefringence. Given the high levels of monoclonal light chains (Bence-Jones proteins) found in this patient's urine, which type of amyloid protein is most likely deposited in his tissues?

- a) AA
- b) AL
- c) Abeta
- d) ATTR protein

Q23. A 2-year-old boy is brought to the pediatrician due to a persistent, itchy, scaly rash in the flexural folds of his elbows and knees. His mother reports that he has a history of prolonged bleeding from the umbilical stump at birth and frequently develops large bruises and "tiny red spots" (petechiae) over his trunk, even with minimal trauma. Additionally, the child has been hospitalized twice in the last year for *Streptococcus pneumoniae* sepsis and recurrent ear infections. Laboratory studies show a low platelet count with a significantly decreased mean platelet volume (very small platelets). Flow cytometry reveals a mutation in a protein responsible for actin cytoskeleton remodeling in hematopoietic cells. Which of the following is the most likely diagnosis?



- a) Chédiak-Higashi syndrome
- b) Wiskott-Aldrich syndrome
- c) Leukocyte Adhesion Deficiency
- d) Hyper-IgE syndrome

Q24. A 62-year-old male presents with a rapidly growing, firm mass in his right thigh. A core needle biopsy is performed. Microscopic examination reveals cells that show a total lack of structural and functional differentiation, resembling no known adult tissue. The cells are highly pleomorphic with a high nuclear-to-cytoplasmic (N:C) ratio, hyperchromatic nuclei, and numerous tripolar mitotic figures. The pathologist notes that the tumor is "primitive" in appearance. Which of the following terms best describes this lack of differentiation in a malignant neoplasm?

- a) Metaplasia
- b) Dysplasia
- c) Anaplasia
- d) Desmoplasia

Q25. A 50-year-old patient undergoes a colonoscopy, which reveals a large sessile polyp in the sigmoid colon. Molecular profiling of the tissue reveals a loss-of-function mutation in a key tumor suppressor protein. In its active, hypo phosphorylated state, this protein binds to the E2F transcription factor, effectively "braking" the cell cycle and preventing the cell from transitioning from the G1 phase to the S phase. Which of the following proteins is responsible for this regulatory "checkpoint"?

- a) p53
- b) RB protein
- c) MYC
- d) BCL-2



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Q26. A tumor cell line demonstrates overexpression of cyclin D, leading to uncontrolled cell cycle progression. Which phase of the cell cycle is primarily regulated by cyclin D?

- a) G0 phase
- b) G1 phase
- c) S phase
- d) M phase

Q27. A 42-year-old male is undergoing intensive total body irradiation (TBI) as part of a conditioning regimen prior to a hematopoietic stem cell transplant for refractory leukemia. Within hours of the first dose, he experiences a dramatic drop in his peripheral lymphocyte count, far preceding any changes in his red blood cells or platelets. Based on the principles of radiation biology, which of the following cell types is the least radioresistant?

- a) Mature neurons
- b) Cardiac myocytes
- c) Lymphocytes
- d) Osteocytes

Q28. A 19-year-old male recently diagnosed with Burkitt Lymphoma begins his first cycle of aggressive induction chemotherapy. Twelve hours after the administration of cytotoxic drugs, the patient develops muscle cramps, cardiac palpitations, and decreased urine output. Laboratory investigations reveal a significant spike in Serum Uric Acid (12.2 mg/dL) and Potassium (6.5 mEq/L). His Calcium is low (6.1 mg/dL), and his Phosphorus is markedly elevated. The patient's ECG shows peaked T-waves. Which of the following describes the most likely underlying pathophysiology and the specific renal complication expected in this patient?

- a) Hypercalcemia due to bone metastasis; Renal stones
- b) Hyperuricemia due to nucleic acid breakdown; Uric acid nephropathy
- c) Hyponatremia due to SIADH; Acute tubular necrosis
- d) Hypokalemia due to vomiting; Interstitial nephritis

Q29. A 58-year-old male presents with a rapidly enlarging, pigmented lesion on his upper back. The lesion has irregular borders, variegated colors ranging from dark brown to jet black, and a diameter of 9 mm. A wide local excision is performed. Histopathology reveals nests of atypical melanocytes with prominent "cherry red" nucleoli invading the papillary dermis. To confirm the diagnosis and differentiate it from other spindle-cell or poorly differentiated tumors, the pathologist performs immunohistochemical staining. The tumor cells show a highly specific marker involved in the maturation of melanosomes. Which of the following is the most specific immunohistochemical stain used for the diagnosis of this condition?

- a) Cytokeratin
- b) Vimentin
- c) HMB-45
- d) Desmin

Q30. A 28-year-old woman presents to the genetic clinic following her third consecutive first-trimester miscarriage. Her family history is notable for a younger brother with Down Syndrome. A peripheral blood karyotype is performed on the woman, which reveals a total of 45 chromosomes. Detailed analysis shows that the long arms of two acrocentric chromosomes have fused at the centromere, with the loss of their respective short arms. Which of the following describes this specific chromosomal abnormality and the most likely pair of chromosomes involved?

- a) Reciprocal translocation; Chromosomes 9 and 22
- b) Robertsonian translocation; Chromosomes 14 and 21
- c) Pericentric inversion; Chromosome 16
- d) Isochromosome formation; X chromosome



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Q31. A 7-year-old child presents with severe obesity, hyperphagia, short stature, small hands and feet, and hypogonadism. Genetic analysis reveals absence of expression of paternally active genes in the 15q11-q13 region. Further testing shows two maternal copies of chromosome 15 with no paternal contribution. Which diagnosis best fits these findings?

- a) Angelman syndrome due to maternal deletion of 15q11-q13
- b) Prader-Willi syndrome caused by paternal deletion of 15q11-q13
- c) Prader-Willi syndrome caused by maternal uniparental disomy of chromosome 15
- d) Angelman syndrome caused by paternal uniparental disomy of chromosome 15

Q32. A 4-year-old boy is brought to a developmental pediatrician due to a profound delay in speech development; he is currently non-verbal and communicates only through basic gestures. His parents describe him as having an "unusually happy disposition," noting that he frequently bursts into paroxysms of inappropriate laughter even when there is no apparent stimulus. On physical examination, the child is microcephalic and exhibits a wide-based, "jerky" gait with his arms held in a flexed, uplifted position while walking. He has a prominent chin (prognathism) and a wide mouth with spaced teeth. The mother mentions that he has a fascination with water and suffers from frequent seizures. Genetic testing reveals a deletion on the long arm of Chromosome 15, specifically involving the maternal copy. Which of the following is the most likely diagnosis, and what is the underlying genetic mechanism?

- a) Prader-Willi Syndrome; Maternal Imprinting (Paternal deletion)
- b) Angelman Syndrome; Paternal Imprinting (Maternal deletion)
- c) Fragile X Syndrome; Trinucleotide repeat expansion
- d) Cri-du-chat Syndrome; Deletion of 5p arm

Q33. A 24-year-old male, who is 6 feet 7 inches tall, presents to the cardiology clinic for a routine evaluation. He has a history of high-grade myopia and recently experienced a sudden "blurring" of vision in his left eye. On physical examination, the physician notes several distinct features: he has exceptionally long, slender fingers (arachnodactyly), and a positive Steinberg sign (the thumb projects beyond the ulnar border of the clenched fist). Further examination reveals a high-arched palate and a chest wall deformity known as pectus excavatum. Slit-lamp examination confirms ectopia lentis (dislocation of the lens), specifically in an upward and outward direction. Auscultation of the heart reveals a mid-systolic click followed by a late systolic murmur, and an echocardiogram shows dilation of the ascending aorta. Given this constellation of clinical findings, which of the following genes is most likely mutated in this patient?

- a) COL1A1
- b) FBN1
- c) TGFBR1
- d) FBN2

Q34. A couple is trying to have a child for past 2 years. After unsuccessful trials they went for respective examinations. The investigations for female came out to be absolutely normal but male shows presence of (a) gynaecomastia, (b) scanty pubic hairs, (c) atrophied testes. Comment on the diagnosis:

- a) Turner syndrome
- b) Downs syndrome
- c) Klinefelter's syndrome
- d) Edward syndrome

Q35. A 4-year-old boy is brought to the clinic for a routine check-up. On physical examination, the physician notes a short stature, a webbed neck, and low-set ears. Evaluation of the heart reveals a systolic murmur, and an echocardiogram confirms Pulmonary Stenosis. Genetic testing is ordered due to the striking resemblance to Turner Syndrome, but the patient has a normal 46,XY karyotype. Which of the following is the most likely genetic defect associated with this condition?



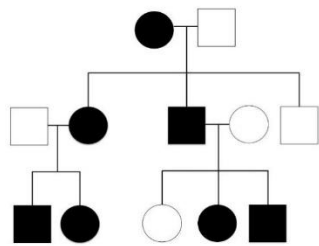
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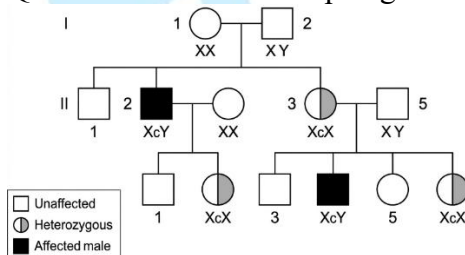
- a) Mutation in the FBN1 gene
- b) Mutation in the PTPN1
- c) 45,XO Karyotype
- d) 47, XXY Karyotype

Q36. Comment on the pedigree analysis:



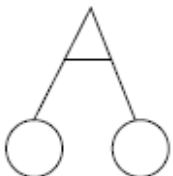
- a) AR
- b) AD
- c) XLR
- d) XLD

Q37. Comment on the pedigree analysis:



- a) AR
- b) AD
- c) XLR
- d) XLD

Q38. Recognise the symbol used for:



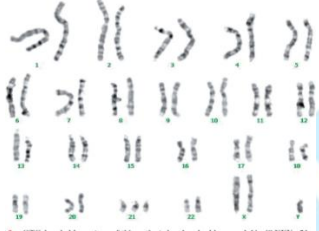


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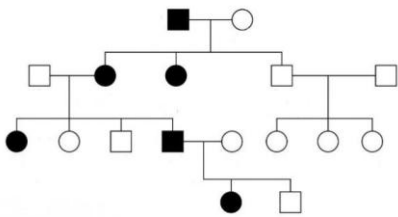
- a) Dizygotic twins
- b) Monozygotic twins
- c) Consanguinity
- d) Pregnancy

Q39. Identify the karyotyping image :



- a) Patau Syndrome
- b) Edward Syndrome
- c) Down Syndrome
- d) Turner Syndrome

Q40. Comment on the pedigree analysis:



- a) AR
- b) AD
- c) XLR
- d) XLD

Q41. A 32-year-old woman presents to the genetic counselor because she is planning a pregnancy. She is healthy but mentions that her father has recently developed an intention tremor and cerebellar ataxia. She has a younger brother with intellectual disability, a long face, and large ears. Genetic testing reveals that the woman has 85 CGG repeats in the 5' untranslated region (UTR) of the FMR1 gene on the X chromosome. She is informed that while she is currently asymptomatic, her children—particularly her sons—are at a significantly higher risk of a more severe clinical presentation compared to her brother or father. Which of the following describes the genetic mechanism responsible for the increasing severity and earlier onset of this condition in successive generations?

- a) Locus Heterogeneity
- b) Pleiotropy
- c) Anticipation
- d) Variable Expressivity

Q42. A 28-year-old female presents to the outpatient department with a 6-month history of progressive fatigue, generalized weakness, and exertional dyspnea. She also mentions that her fingernails have become brittle and oddly shaped. On further questioning, she admits to an unusual craving for eating ice cubes and raw starch (pica). She has a history of heavy menstrual bleeding (menorrhagia) over the last two years.

Laboratory Investigations:

Hemoglobin: 7.5 g/dL (Normal: 12–15)

MCV: 62 fL (Normal: 80–100)



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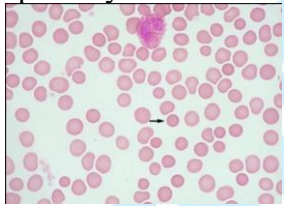
MCHC: 28 g/dL (Low)

Mentzer Index: 14

Which of the following is the most likely diagnosis and the most specific laboratory finding to confirm it?

- a) Megaloblastic Anemia; Increased Homocysteine
- b) Iron Deficiency Anemia (IDA); Low Serum Ferritin
- c) Thalassemia Minor; HbA2 > 3.5%
- d) Anemia of Chronic Disease; Increased Hepcidin

Q43. A patient presents with jaundice, splenomegaly, and elevated reticulocyte count. Peripheral smear shows spherocytes. Which condition is most likely responsible?



- a) Iron deficiency anemia
- b) Immune hemolytic Anemia
- c) Hereditary spherocytosis
- d) Thalassemia

Q44. A 26-year-old male medical student presents to the emergency department with sudden onset of yellowish discoloration of the eyes and dark, "cola-colored" urine. He was recently prescribed a course of Primaquine for the radical cure of Plasmodium vivax malaria. He also recalls a similar, milder episode in childhood after consuming a meal containing fava beans. Supravital Stain (Crystal Violet) Reveals dark, round inclusions within the RBCs. Which of the following is the most likely enzymatic defect responsible for this patient's acute hemolytic episode?

- a) Pyruvate Kinase deficiency
- b) Glucose-6-Phosphate Dehydrogenase
- c) Glucokinase deficiency
- d) Glutathione Reductase deficiency

Q45. A 34-year-old male presents with severe, cramping abdominal pain and a history of passing dark-red urine, particularly in the morning's first void. He has been evaluated previously for mild pancytopenia. Over the last 48 hours, the abdominal pain has intensified, and physical examination now reveals a tender, enlarged liver and new-onset ascites. Doppler Ultrasound: Shows total occlusion of the hepatic veins by a thrombus. Flow cytometry of his blood shows a deficiency in CD55 (DAF) and CD59 (MIRL) on the surface of his red blood cells, making them susceptible to complement-mediated lysis. What is the name of this characteristic thrombotic complication involving the hepatic veins seen in this patient?

- a) Portal vein thrombosis
- b) Budd–Chiari syndrome
- c) Hepatic Veno-occlusive disease
- d) Splenic vein thrombosis

Q46. A 30-year-old woman is referred to the hematology clinic due to a 3-week history of worsening fatigue, mild jaundice (icterus), and dark-colored urine. She also mentions a history of joint pains and a "butterfly rash" on her face during the summer months.

Laboratory Investigations:

Hemoglobin: 7.2 g/dL

Reticulocyte Count: 8%



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Bilirubin: Elevated Indirect Bilirubin.

Peripheral Smear: Shows numerous spherocytes and some polychromasia.

Direct Coombs: Positive

Which of the following is the most likely diagnosis?

- a) Hereditary Spherocytosis
- b) Autoimmune Hemolytic Anemia
- c) Sick Cell Trait
- d) Microangiopathic Hemolytic Anemia

Q47. A 26-year-old Southeast Asian woman in her 28th week of pregnancy undergoes a routine ultrasound. The scan reveals severe generalized fetal edema (fetal hydrops), including pleural effusion and ascites. There is significant placentomegaly. Sadly, fetal demise occurs shortly after. Post-mortem Hemoglobin Analysis The fetal blood reveals a total absence of Hemoglobin F and Hemoglobin A. Instead, there is a predominance of gamma tetramers. Which of the following genetic defects is responsible for this condition?

- a) Deletion of 2 alpha genes
- b) Deletion of 3 alpha genes
- c) Deletion of 4 alpha genes
- d) Deletion of 1 alpha genes

Q48. A 28-year-old female primary school teacher presents with a 4-month history of progressive fatigue and "breathlessness" when climbing stairs. Her husband notes that she has started chewing on large amounts of ice throughout the day (pagophagia). She has a history of heavy menstrual cycles lasting 7–8 days. Skin/Nails: The nails are thin, brittle, and show a distinct concave, spoon-shaped appearance (koilonychia).

Laboratory Results:

Hb: 8.2 g/dL

MCV: 68 fL

Mentzer Index: 15.2

Iron Profile: Low Serum Iron, Low Transferrin Saturation.

Which of the following laboratory parameters is the most specific and earliest indicator that her total body iron stores are exhausted?

- a) Reduced Hemoglobin concentration
- b) Decreased Mean Corpuscular Volume
- c) Decreased Serum Ferritin levels
- d) Increased Total Iron Binding Capacity

Q49. A 55-year-old male with a 12-year history of poorly controlled Rheumatoid Arthritis presents for a follow-up. He reports chronic "low energy" but denies pica, melena, or any recent change in diet. He is currently on disease-modifying anti-rheumatic drugs (DMARDs) and has active joint swelling in his wrists and MCP joints.

Laboratory Results:

Hb: 10.4 g/dL

MCV: 82 fL

Serum Iron: Low

TIBC: Low

Serum Ferritin: 350 ng/mL

ESR/CRP: Significantly elevated.

Which of the following physiological mechanisms, driven by the inflammatory cytokine IL-6, is primarily responsible for the low serum iron seen in this patient?

- a) Chronic occult blood loss from the GI tract due to NSAID use
- b) Increased synthesis of Hepcidin leading to iron sequestration in macrophages



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- c) Direct suppression of the bone marrow by autoantibodies
- d) Decreased intestinal absorption of Vitamin B12

Q50. A patient with chronic tuberculosis presents with anemia. Which of the following laboratory profiles is most consistent with anemia of chronic disease?

- a) ↓ Iron, ↑ TIBC, ↓ Ferritin
- b) ↓ Iron, ↓ TIBC, ↑ Ferritin
- c) ↑ Iron, ↓ TIBC, ↑ Ferritin
- d) ↓ Iron, ↑ TIBC, ↑ Ferritin

Q51. A 32-year-old female presents with progressive fatigue. Laboratory findings reveal severe anemia (Hb 6 g/dL), normal WBC and platelet counts, and markedly reduced reticulocyte count. Bone marrow shows absence of erythroid precursors with preserved myeloid and megakaryocytic lineages. What is the most likely diagnosis?

- a) Aplastic anemia
- b) Pure red cell aplasia
- c) Myelodysplastic syndrome
- d) Iron deficiency anemia

Q52. A 62-year-old male presents with a two-month history of worsening fatigue and a "burning sensation" on his tongue. He has a history of heavy alcohol consumption and a diet poor in leafy green vegetables. Hemoglobin: 9.2g/dL MCV: 118 fL (Marked Macrocytosis) , Peripheral Smear: Reveals large, oval-shaped red blood cells (macro-ovalocytes) and several neutrophils with 6 to 8 nuclear lobes. What is the primary cellular mechanism responsible for the morphological findings seen on this patient's peripheral blood smear?

- a) Defective mitochondrial heme synthesis
- b) Impaired DNA synthesis leading to delayed nuclear maturation
- c) Premature destruction of RBCs in the splenic sinusoids
- d) Cytoplasmic iron overload within the mitochondria

Q53. A 55-year-old woman is brought to the clinic by her daughter due to increasing "clumsiness" and frequent falls. The patient also complains of a "pins and needles" sensation in her hands and feet. Medical history is significant for Atrophic Gastritis and a diagnosis of Pernicious Anemia. Neurological: Loss of vibration and position (proprioception) sense in the lower limbs. Reflexes: Positive Babinski sign and absent ankle jerks. Gait: Sensory ataxia with a positive Romberg sign. While both Folate and Vitamin B12 deficiency cause an increase in serum homocysteine, which specific biochemical accumulation is unique to this patient's condition and directly responsible for the demyelination of the spinal cord?

- a) Elevated Serum Folate
- b) Increased Methylmalonic Acid (MMA)
- c) Decreased Serum Ferritin
- d) Increased Protoporphyrin

Q54. A 34-year-old male presents with a 4-month history of fatigue and recurrent episodes of dark-colored urine, which he notices primarily in his first morning void. He also reports recent onset of severe abdominal pain. Flow Cytometry: Shows a distinct population of red blood cells and granulocytes lacking CD55 (Decay-Accelerating Factor) and CD59 (MAC-Inhibitory Protein). The patient is started on a monoclonal antibody that binds to the C5 complement protein, preventing the formation of the Membrane Attack Complex (MAC) and reducing intravascular hemolysis. Which of the following is the drug of choice for this patient?

- a) Rituximab
- b) Eculizumab
- c) Infliximab



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d) Abciximab

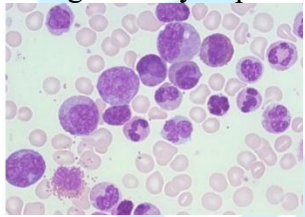
Q55. A 14-year-old boy with a known history of Sickle Cell Disease (HbSS) is brought to the emergency department with a 6-hour history of excruciating pain in his lower back and bilateral thighs. He was on a school hiking trip in a high-altitude mountainous region when the symptoms began. On physical examination, he is in visible distress, tachycardic (Pulse 110/min), and has mild scleral icterus. His oxygen saturation is 90% on room air. Peripheral Smear Numerous sickle-shaped erythrocytes and Howell-Jolly bodies. The physician explains that the environmental conditions at high altitude triggered the deoxygenation of his hemoglobin, leading to the polymerization of hemoglobin S and subsequent "sickling" of his red blood cells, causing a Vaso-occlusive Crisis. What should be the next most appropriate step?

- a) Saline infusion
- b) O₂ supplementation
- c) Hydroxyurea
- d) Exchange transfusion

Q56. A 2-year-old child presents with progressive pallor, failure to thrive, and recurrent transfusion requirement since infancy. On examination, there is frontal bossing and hepatosplenomegaly. Peripheral smear shows microcytic hypochromic anemia with target cells and nucleated RBCs. What is the most likely diagnosis?

- a) Iron deficiency anemia
- b) β -thalassemia major
- c) Sideroblastic anemia
- d) Anemia of chronic disease

Q57. 47-year-old male presents with early satiety, fatigue, and abdominal fullness. On examination, there is massive splenomegaly. Laboratory findings reveal hemoglobin of 10 g/dL and a total leukocyte count of 120,000/cmm. Peripheral smear shows a spectrum of granulocytic cells including myeloblasts, promyelocytes, myelocytes, and metamyelocytes. Basophils are mildly increased. Bone marrow examination reveals hypercellularity with granulocytic proliferation. What is the most likely diagnosis?



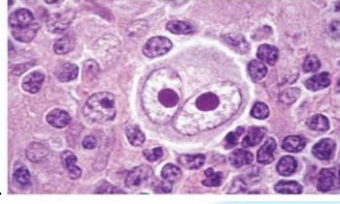
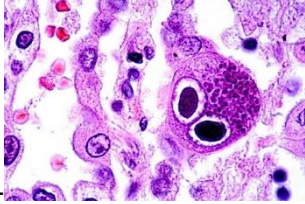
- a) Acute Myeloid Leukemia
- b) Chronic Myeloid Leukemia
- c) Acute Lymphoblastic Leukemia
- d) Chronic Lymphocytic Leukemia

Q58. A patient with CML develops worsening anemia, thrombocytopenia, and >20% blasts in bone marrow. What phase of disease does this represent?

- a) Chronic phase
- b) Accelerated phase
- c) Blast crisis
- d) Remission phase

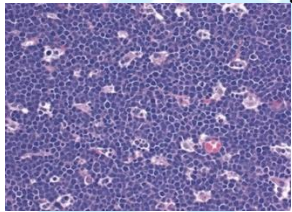
Q59. Identify the type of cell:

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- A- Hodgkins's lymphoma B- Non-Hodgkins lymphoma
 a) A- Hodgkins's lymphoma B- Cytomegalovirus
 b) A- non-Hodgkins's lymphoma B- Cytomegalovirus
 c) A- Cytomegalovirus B- Hodgkins's lymphoma

Q60. A 9-year-old boy from an endemic region presents with rapidly enlarging swelling over the jaw associated with fever and weight loss. On examination, a firm mass is noted in the maxillary region. Histopathological examination reveals a "starry-sky" appearance due to macrophages interspersed among proliferating lymphoid cells. The tumor cells show a translocation involving chromosomes 8 and 14, leading to dysregulated gene expression. Which of the following best describes the primary pathogenic mechanism involved in this disease?



- a) Chromosomal translocation
 b) Viral genome integration into host DNA
 c) Tumor suppressor gene deletion
 d) Direct activation of oncogene c-MYC overexpression

Q61. A 35-year-old male is rushed to the emergency department with profuse bleeding from his gums and multiple spontaneous bruises (ecchymoses) over his trunk. His laboratory results reveal a "panic value" of pancytopenia (Hb: 7.2 g/dL, WBC: $1.5 \times 10^3/\mu\text{L}$, Platelets: $12,000/\mu\text{L}$). A peripheral smear is performed immediately. Smear: Shows numerous large "blasts" with heavy granulation and multiple needle-like inclusions known as Auer rods (specifically "faggot cells"). Coagulation Profile: Prolonged PT/APTT and decreased Fibrinogen, suggesting DIC. Cytogenetics: Confirms a translocation between the long arms of chromosomes 15 and 17 [t(15;17)]. What is the most likely diagnosis, and what is the target of the standard therapy used to "differentiate" these cells?

- a) AML-M0; CD34
 b) Acute Promyelocytic Leukemia; Retinoic Acid Receptor
 c) Acute Monocytic Leukemia; CD14
 d) ALL; TdT

Q62. A 60-year-old bank manager presents for a routine physical. He mentions feeling "full" after eating small meals (early satiety) and having a dull ache in his left upper abdomen. On examination, his spleen is massive, crossing the midline. WBC Count: $180,000/\mu\text{L}$ (Extreme Leukocytosis). Peripheral Smear: Shows a "sea of cells" including neutrophils, bands, metamyelocytes, myelocytes, and promyelocytes (all stages of maturation). There is a notable Basophilia. LAP Score: Significantly decreased. Cytogenetics: Reveals the Philadelphia Chromosome, a result of the t(9;22) translocation creating the BCR-ABL fusion gene. Which of the following is the most likely diagnosis?

- a) Acute Myeloid Leukemia
 b) Leukemoid Reaction
 c) Chronic Myeloid Leukemia

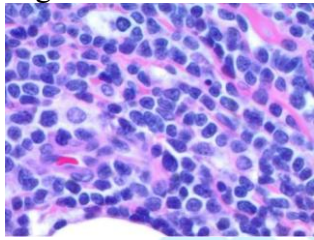


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d) Polycythemia Vera

Q63. A 5-year-old boy is brought to the pediatrician because he refuses to walk due to "leg pain." His mother notes he has had a persistent fever and "lumps" in his neck. On examination, he has generalized lymphadenopathy and hepatosplenomegaly. WBC Count: Elevated with 80% small, uniform cells with scant cytoplasm and "punched out" nucleoli (lymphoblasts) Bone Marrow: Replaced by >20% blasts. Special Stains: Myeloperoxidase (MPO) is negative. Which of the following markers is most specific for identifying these cells as "blasts" of lymphoid origin?



- a) CD13 and CD33
- b) Terminal Deoxynucleotidyl Transferase
- c) CD117 (c-kit)
- d) Myeloperoxidase

Q64. A 28-year-old female presents with fatigue, bleeding gums, and easy bruising. Peripheral smear shows blasts with multiple Auer rods. Cytogenetic analysis reveals t(15;17). What is the most likely complication associated with this condition?

- a) Tumor lysis syndrome
- b) Disseminated intravascular coagulation (DIC)
- c) Hyperviscosity syndrome
- d) Splenic rupture

Q65. A 48-year-old male banker presents for a check-up due to increasing fatigue, unintentional weight loss of 6 kg over two months, and a "dragging sensation" in his left upper abdomen. On physical examination, the physician palpates a firm, non-tender mass extending 10 cm below the left costal margin towards the iliac crest (massive splenomegaly). Cytogenetics: Karyotyping reveals a shortened chromosome 22, known as the Philadelphia chromosome. The translocation involved in this condition results in the formation of a chimeric gene that encodes a protein with constitutive tyrosine kinase activity. What is this specific molecular abnormality?

- a) Sequestration of retinoic acid receptor by the PML-RARA fusion
- b) BCR-ABL fusion gene on chromosome 22q
- c) Overexpression of BCL-2 due to t(14;18)
- d) Transcriptional activation of c-MYC on chromosome 8

Q66. A 52-year-old male presents with a very high white blood cell count of 60,000/ μ L. The peripheral smear shows a "left shift" with many immature myeloid cells (myelocytes and metamyelocytes). The clinician is trying to distinguish between Chronic Myeloid Leukemia (CML) and a Leukemoid Reaction (secondary to a severe underlying infection). Which of the following laboratory findings in this patient is most characteristic of CML and helps rule out a Leukemoid Reaction?

- a) Significant increase in the Total Leucocyte Count
- b) Presence of myelocytes and metamyelocytes on the smear
- c) A low Leukocyte Alkaline Phosphatase score
- d) Presence of a high percentage of mature neutrophils



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- Q67. A 55-year-old male presents with headache, dizziness, and generalized pruritus after a hot bath. Examination reveals facial plethora and splenomegaly. Laboratory findings show increased Hb, hematocrit, and RBC mass with low erythropoietin levels. What is the most likely diagnosis?
- Secondary polycythemia
 - Relative polycythemia
 - Polycythemia vera
 - Chronic myeloid leukemia
- Q68. A 60-year-old male presents with a 6-month history of painless, waxing and waning enlargement of lymph nodes in his neck and armpits. He denies fever, night sweats, or weight loss. On examination, he has generalized, non-tender, rubbery lymphadenopathy involving the cervical and axillary regions. Histopathology: Shows a "nodular" or follicular pattern of growth consisting of small cleaved cells (centrocytes) and larger non-cleaved cells (centroblasts. Immunohistochemistry: The neoplastic follicles are positive for BCL-2 protein (unlike reactive follicles, which are BCL-2 negative). Cytogenetics: Reveals a reciprocal translocation between the immunoglobulin heavy chain locus on chromosome 14 and the BCL-2 gene on chromosome 18 [t(14;18)]. Which of the following is the most likely diagnosis?
- Burkitt Lymphoma (t[8;14])
 - Follicular Lymphoma
 - Mantle Cell Lymphoma (t[11;14])
 - Diffuse Large B-Cell Lymphoma (DLBCL)
- Q69. A 52-year-old female presents with frequent headaches, dizziness, and a distressing "burning" sensation in her hands and feet associated with reddish-blue discoloration (erythromelalgia). She has no history of smoking, chronic infections, or recent surgery. On examination, her spleen is palpable 2 cm below the left costal margin. Platelet Count: 9,80,000/ μ L (Markedly elevated). Bone Marrow Biopsy: Shows hypercellularity with clusters of enlarged, "staghorn" megakaryocytes. Molecular Testing: Positive for the JAK2 V617F mutation. After ruling out reactive causes of high platelets (like iron deficiency or inflammation), what is the most likely diagnosis?
- Polycythemia Vera
 - Essential Thrombocythemia (ET)
 - Chronic Myeloid Leukemia (CML)
 - Primary Myelofibrosis (PMF)
- Q70. A 60-year-old male presents with generalized lymphadenopathy and weight loss. Lymph node biopsy shows diffuse proliferation of large atypical B cells. What is the most likely diagnosis?
- Follicular lymphoma
 - Burkitt lymphoma
 - Diffuse large B-cell lymphoma (DLBCL)
 - Mantle cell lymphoma
- Q71. A 65-year-old retired farmer presents to the orthopedics clinic with persistent, dull aching pain in his lower back and ribs that worsens with movement. He also reports significant fatigue and two episodes of pneumonia in the last six months. Hemoglobin: 9.0 g/dL (Normocytic normochromic anemia). Hypercalcemia, Serum Creatinine: 2.1 mg/dL Urinalysis: Negative for glucose and protein on dipstick, but Sulfosalicylic acid test is positive. Peripheral Smear: RBCs are seen stacked like coins (Rouleaux formation). Radiology image is shown below. Which of the following is the most likely diagnosis?



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- a) Osteoblastic Metastatic Prostatic Carcinoma
- b) Multiple Myeloma
- c) Senile Osteoporosis
- d) Paget's Disease of Bone

Q72. A patient presents with hemarthrosis and deep tissue bleeding. Laboratory findings show prolonged aPTT with normal PT and bleeding time. What is the most likely diagnosis?



- a) von Willebrand disease
- b) Hemophilia A
- c) Platelet disorder
- d) Vitamin K deficiency

Q73. A patient presents with prolonged PT and normal aPTT. Which clotting factor deficiency is most likely?

- a) Factor VIII
- b) Factor IX
- c) Factor VII
- d) Factor X

Q74. A 24-year-old female presents with petechiae, epistaxis, and easy bruising. Laboratory findings show platelet count of 25,000/ μ L with normal PT and aPTT. Bone marrow shows increased megakaryocytes. What is the most likely diagnosis?

- a) Aplastic anemia
- b) Disseminated intravascular coagulation
- c) Immune thrombocytopenic purpura
- d) Von willibrand disease

Q75. A 4-year-old child presents with diarrhea followed by pallor, reduced urine output, and petechiae. Laboratory findings show anemia, thrombocytopenia, and elevated creatinine. Peripheral smear shows schistocytes. What is the most likely diagnosis?

- a) Immune thrombocytopenic purpura
- b) Disseminated intravascular coagulation
- c) Hemolytic uremic syndrome
- d) Hemophilia



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Q76. A 10-year-old boy has mucosal bleeding. Platelet count is slightly reduced, bleeding time is prolonged, and platelets appear large on smear. PT, apTT is normal. Platelet aggregation is absent with ristocetin. What is the diagnosis?

- a) Glanzmann thrombasthenia
- b) Bernard-Soulier syndrome
- c) von Willebrand disease
- d) ITP

Q77. A woman presents with menorrhagia and prolonged bleeding after dental extraction. Both bleeding time and aPTT are prolonged. PT is normal. Which is the most likely diagnosis?

- a) DIC
- b) Hemophilia B
- c) von Willebrand disease
- d) Vitamin K deficiency

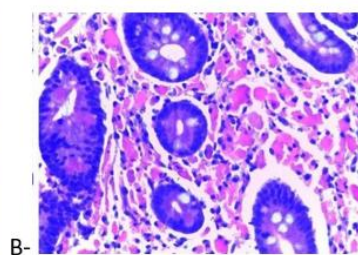
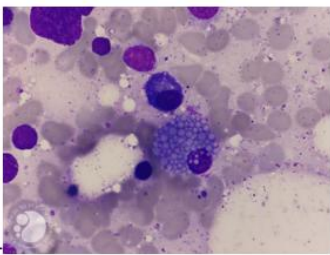
Q78. A 5-year-old girl is brought to the pediatric emergency department due to a sudden decrease in urine output and noticeable "puffiness" of her face and eyelids. Her mother is extremely concerned because the child has become "pale as a ghost" and has developed unexplained bruising on her shins. The mother notes that the child had a severe bout of bloody diarrhea and abdominal cramps following a family cookout eight days ago.

Skin: Multiple petechiae and ecchymoses are scattered across the lower extremities. Hemoglobin: 6.8 g/dL with a Coombs-negative hemolytic profile.

Peripheral Smear: Numerous fragmented erythrocytes, specifically schistocytes and helmet cells, are seen against a background of reduced platelets. Based on the integration of these findings, what is the most likely diagnosis?

- a) Disseminated Intravascular Coagulation (DIC)
- b) Hemolytic Uremic Syndrome (HUS)
- c) Immune Thrombocytopenic Purpura (ITP)
- d) Acute Post-Streptococcal Glomerulonephritis (APSGN)

Q79. Identify the type of cell given in the image:



- a) A-Dutcher body B- Flame cell
- b) A- Mott Cell B- Dutcher body
- c) A- Mott Cell B- Russell body
- d) A- Russell Body B- Dutcher body

Q80. A 58-year-old male presents with severe retrosternal chest pain radiating to the left arm for 2 hours. ECG shows ST elevation. Which of the following is the earliest biochemical marker to rise in myocardial infarction?

- a) CK-MB
- b) Troponin I
- c) Myoglobin
- d) LDH



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Q81. A 30-year-old Asian female presents to the vascular clinic complaining of easy fatiguability of her arms while hanging laundry and occasional "dizziness" when she turns her head. On physical examination, the physician notes that the radial pulse is nearly absent on the right side and significantly weakened on the left. Blood Pressure (Left Arm): 150/90 mmHg. Blood Pressure (Right Arm): 100/70 mmHg. Bruit: A systolic bruit is heard over the left subclavian artery. Imaging (MRA): Reveals granulomatous thickening and narrowing of the aortic arch and its major branches. What is the most likely diagnosis?

- a) Giant Cell Arteritis (Temporal Arteritis)
- b) Takayasu Arteritis
- c) Polyarteritis Nodosa (PAN)
- d) Thromboangiitis Obliterans (Buerger's Disease)

Q82. A 4-year-old boy is brought to the pediatric emergency department with a high-grade fever that has lasted for 6 days and is unresponsive to paracetamol. His mother is alarmed because his eyes are very red, though there is no discharge. Oral Cavity: Bright red, swollen lips and a "strawberry tongue" (prominent papillae). Extremities: Marked edema of the hands and feet with beginning desquamation (peeling) of the fingertips. Skin: A polymorphous rash is noted on the trunk, along with a single, enlarged cervical lymph node. Which of the following is the most dreaded complication of this disease, requiring mandatory screening with an Echocardiogram?

- a) Glomerulonephritis
- b) Coronary Artery Aneurysm
- c) Alveolar Hemorrhage
- d) Saddle Nose Deformity

Q83. A 6-year-old boy is brought in for evaluation of gait instability, frequent falls, and early-morning vomiting. Examination shows truncal ataxia and papilledema. MRI reveals a poorly demarcated midline mass arising from the cerebellar vermis, causing effacement of the fourth ventricle. CSF cytology later shows malignant cells. Histopathology demonstrates hyperchromatic small round blue cells with scant cytoplasm, arranged in Homer-Wright rosettes. Which diagnosis best fits these findings?

- a) Primitive neuroectodermal tumor of the cerebellar vermis with CSF dissemination tendency
- b) Fourth-ventricular tumor with perivascular pseudorosettes and true ependymal canals
- c) Pilocytic astrocytoma characterized by biphasic pattern and Rosenthal fibers
- d) Craniopharyngioma with calcifications

Q84. A 55-year-old man presents with progressive unilateral hearing loss, tinnitus, and episodes of imbalance for 8 months. Neurologic exam shows decreased corneal reflex on the affected side. MRI reveals a well-circumscribed enhancing mass at the cerebellopontine angle, extending into the internal acoustic meatus. What is the most likely diagnosis?

- a) Benign nerve sheath tumor arising from Schwann cells of CN VIII, often presenting with hearing loss and CP-angle mass
- b) Dural-based tumor with psammoma bodies that typically attaches to the tentorium or falx
- c) Infiltrative primary brain tumor derived from glial cells, usually located within brain parenchyma rather than CP angle
- d) Tumor arising from ependymal lining of the ventricles, commonly located near the fourth ventricle in children

Q85. A 40-year-old male with seizures has a tumor in the frontal lobe with calcification. Histology shows "fried egg" cells and a delicate capillary network. What is the likely diagnosis?

- a) Oligodendroglioma
- b) Ependymoma
- c) Astrocytoma
- d) Glioblastoma



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Q86. A 2-year-old child presents with hydrocephalus. MRI reveals a tumor in the fourth ventricle. Histology shows perivascular pseudorosettes. What is the likely diagnosis?

- a) Ependymoma
- b) Medulloblastoma
- c) Astrocytoma
- d) Craniopharyngioma

Q87. A 55-year-old man presents with chronic cough. Biopsy shows destruction of alveolar walls with enlarged air spaces without fibrosis. What is the most likely diagnosis?

- a) Chronic bronchitis
- b) Centriacinar emphysema
- c) Panacinar emphysema
- d) Bronchiectasis

Q88. A lung biopsy reveals intra-alveolar hyaline membranes and alveolar collapse. Which of the following best describes the condition?

- a) Chronic bronchitis
- b) Bronchopneumonia
- c) Diffuse alveolar damage
- d) Organizing pneumonia

Q89. A 56-year-old chronic smoker presents with a long history of productive cough lasting more than 3 months each year for the past 2 years. Physical examination reveals coarse crackles, and chest X-ray shows increased bronchovascular markings. Bronchial biopsy demonstrates marked mucous gland hypertrophy. Which measurement is used to assess the severity of this gland enlargement?

- a) Kerley index used to assess interstitial edema in heart failure
- b) Reid index measuring mucous gland thickness relative to bronchial wall thickness
- c) Karnofsky index evaluating functional status in chronic disease
- d) Apgar score assessing neonatal health immediately after birth

Q90. A 64-year-old man is admitted with an acute MI and undergoes stent placement. On day 3, he develops new severe chest pain. Troponin I remains elevated from the initial event. Which cardiac marker is most helpful in detecting reinfarction?

- a) LDH, because it rises quickly and falls within 24 hours
- b) CK-MB, because it's a marker for infarction
- c) Myoglobin, because it remains elevated for 7-10 days
- d) Troponin T, because it normalizes faster than CK-MB

Q91. A 32-year-old woman with known systemic lupus erythematosus presents with fatigue. Echo reveals small, sterile vegetations on both the atrial and ventricular surfaces of the mitral valve leaflets. Which type of endocarditis best explains these findings?

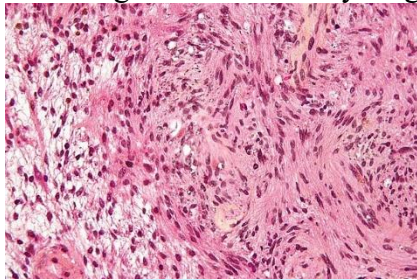
- a) Verrucous endocarditis associated with rheumatic fever, typically forming vegetations along the lines of closure
- b) Nonbacterial thrombotic endocarditis seen in cancer or cachexia, producing bland vegetations along valve edges
- c) Sterile immune-complex-mediated endocarditis of SLE causing vegetations on either side of the valve
- d) Post-streptococcal endocarditis producing verrucae along the line of closure with Aschoff bodies in the myocardium



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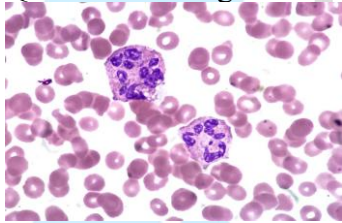
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Q92. A 38-year-old male presents with a two-year history of progressive hearing loss in his right ear and intermittent "ringing" (tinnitus). Recently, he has started experiencing occasional dizziness and a slight loss of sensation on the right side of his face. MRI of the brain reveals a well-circumscribed, 3 cm mass located in the cerebellopontine (CP) angle, originating from the vestibular branch of the 8th cranial nerve. Histopathology: A biopsy of the mass is performed. The pathologist notes two distinct cellular patterns: shown in the image. Which of the following is the most likely diagnosis?



- a) Meningioma
- b) Schwannoma (Acoustic Neuroma)
- c) Glioblastoma Multiforme
- d) Ependymoma

Q93. Following cell demonstrates:



- a) Bone marrow failure
- b) Inclusion bodies in Myeloblasts
- c) Vitamin deficiency
- d) Lyst gene defect

Q94. A 65-year-old male with a 50-pack-year smoking history presents to the pulmonary clinic with a persistent cough, hemoptysis (coughing up blood), and a 10 kg weight loss over the past two months. He also reports recent onset of muscle weakness that improves with repetitive use (Lambert-Eaton Myasthenic Syndrome). A Chest CT reveals a large, hilar (central) mass with extensive mediastinal lymphadenopathy. The bronchial biopsy shows sheets of small, round-to-oval cells with scant cytoplasm, "salt-and-pepper" chromatin, and inconspicuous nucleoli. There is significant evidence of nuclear molding (cells pressing against each other) and high mitotic activity. Which of the following immunohistochemical (IHC) stains would be MOST useful in confirming the nature of this patient's lung tumor?

- a) Cytokeratin 7 (CK7)
- b) Synaptophysin or Chromogranin A
- c) Thyroid Transcription Factor-1 (TTF-1)
- d) p40

Q95. A 12-year-old boy presents with migratory joint pain, a new-onset heart murmur, and purposeless, jerky movements of his hands (Sydenham chorea). His mother notes he had a severe "sore throat" three weeks ago that was not treated. An echocardiogram reveals vegetations along the lines of closure of the mitral valve. If a myocardial biopsy were performed, it would reveal characteristic Aschoff bodies.



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These are areas of focal inflammation consisting of a central zone of fibrinoid necrosis surrounded by a collection of lymphocytes, plasma cells, and large, activated macrophages with "owl-eye" or "caterpillar" nuclei. What are these specialized, activated macrophages called?

- a) Hürthle cells
- b) Anitschkow cells
- c) Reed-Sternberg cells
- d) Langerhans cells

Q96. A 42-year-old male with a known history of mitral valve prolapse presents with a 10-day history of low-grade fever, night sweats, and "shaking chills." He underwent an extensive dental extraction three weeks ago without antibiotic prophylaxis.

Physical Examination:

Vitals: Temperature 38.5 celcius

Cardiovascular: A new-onset high-pitched holosystolic murmur is heard at the apex.

Skin/Eyes: Small, non-tender erythematous macules are noted on his palms (Janeway lesions), and funduscopy exam reveals retinal hemorrhages with pale centers (Roth spots).

Laboratory Investigations:

Blood Cultures: 2 out of 2 sets are positive for Streptococcus viridans.

Echocardiogram (TTE): Reveals a 4 mm mobile mass (vegetation) attached to the atrial surface of the mitral valve

According to the Modified Duke's Criteria, which of the following best describes the diagnostic status of this patient?

- a) Possible Infective Endocarditis (1 Major and 1 Minor)
- b) Definite Infective Endocarditis (2 Major and 3 Minor)
- c) Definite Infective Endocarditis (1 Major and 3 Minor)
- d) Possible Infective Endocarditis (2 Major and 1 Minor)

Q97. A 6-year-old boy is brought to the clinic by his parents because of a "concerning rash" on his legs and buttocks that appeared two days ago. The mother mentions that the child had a cold and sore throat about two weeks ago. Numerous non-blanching, raised, purple-red macules and papules (palpable purpura) are concentrated on the lower extremities and buttocks. The child is limping and complains of pain and swelling in both ankles. Mild, diffuse colicky abdominal pain is noted.

A skin biopsy is performed on one of the lesions. Which of the following immunofluorescence findings is pathognomonic for this condition?



- a) Linear deposition of IgG along the glomerular basement membrane
- b) Granular deposition of IgA and C3 in the walls of small dermal vessels
- c) Ribbon-like deposition of C3 in the basement membrane
- d) "Fish-net" pattern of IgG in the epidermis

Q98. A 45-year-old male presents with a persistent "sinus infection" that has not responded to multiple courses of antibiotics over the last three months. He complains of chronic nasal congestion, occasional bloody nasal discharge, and a noticeable "caving in" of the bridge of his nose. Nasal mucosal ulcerations and a visible saddle-nose deformity. Bilateral scattered crackles are heard. Palpable purpura over the lower limbs.



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Urinalysis: Shows "coke-colored" urine with RBC casts and significant proteinuria. Serum Creatinine: 2.4 mg/dL (Elevated). Chest X-ray Reveals multiple bilateral cavitory nodules. Serology Strongly positive for cytoplasmic antineutrophil cytoplasmic antibodies (c-ANCA).

A biopsy of the lung lesions is performed. Which of the following histological findings would be most characteristic of this patient's condition?

- a) Non-caseating granulomas without vascular involvement
- b) Granulomatous inflammation, geographic necrosis, and vasculitis
- c) Eosinophilic-rich granulomatous inflammation with asthma history
- d) IgA-dominant immune complex deposition in vessel walls

Q99. Which of the following will not be included in diagnosing Multiple Myeloma?

- a) Hypercalcemia
- b) Schistocytes
- c) Monoclonal protein quantity
- d) Less EPO

Q100. Approximately 65 days left for the exam. Lots of things to do and lot of plans I have. Too much anxiety and panic. Which amongst the following would be the best approach moving forward?

- a) Solving as many question as possible and write all the mistakes in a separate notebook.
- b) Keep analysing myself by giving GT every 5th day from now and learn all the PYT's only.
- c) Switching to revision videos and rewatching them again and again and notes should be avoided as they will take a lot of time.
- d) Having a balanced approach wherein I have to understand that something or the other will be left and reading the content and solving MCQ's should both be given due importance wherein knowing the content holds primary importance and then comes remembering and applying it in MCQ.

ANSWERS

1	b
2	b
3	b
4	b
5	c
6	d
7	b
8	c
9	b
10	a
11	b
12	b
13	b
14	d
15	c
16	a
17	b

18	a
19	b
20	b
21	d
22	b
23	b
24	c
25	b
26	b
27	c
28	b
29	c
30	b
31	c
32	b
33	b
34	c

35	b
36	b
37	c
38	b
39	c
40	d
41	c
42	b
43	b
44	b
45	b
46	b
47	c
48	c
49	b
50	b
51	b

52	b
53	b
54	b
55	b
56	b
57	b
58	c
59	d
60	d
61	b
62	c
63	b
64	b
65	b
66	c
67	c
68	b

69	b
70	c
71	b
72	b
73	c
74	c
75	c
76	b
77	c
78	b
79	b
80	c
81	b
82	b
83	a
84	a
85	a



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86	a
87	b
88	c

89	b
90	b
91	c

92	b
93	c
94	b

95	b
96	b
97	b

98	b
99	b
100	d

